



Computer Vision and Image Processing

Lecture 21

Vision Transformers

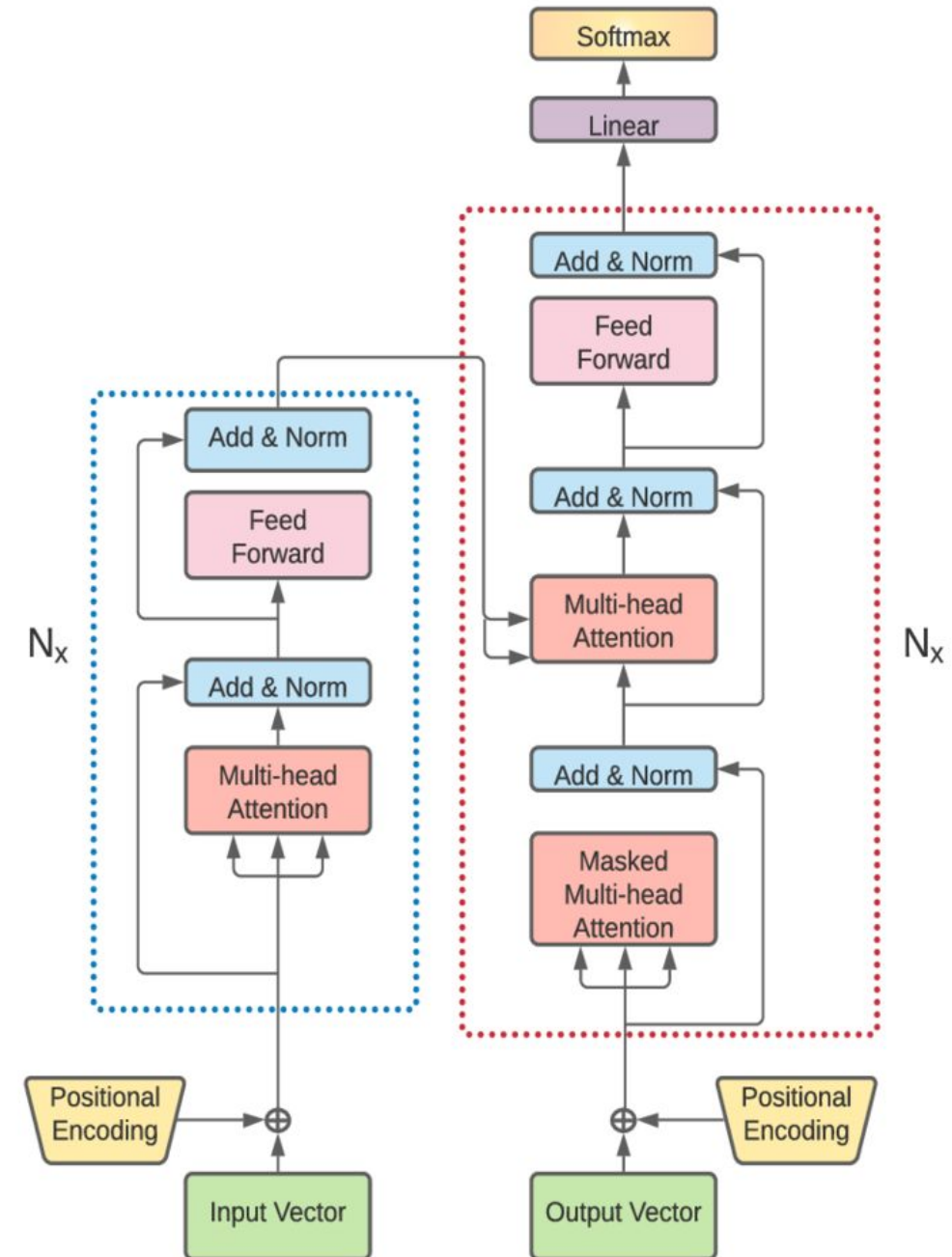
What are Transformers?

- A deep learning architecture originally developed for Natural Language Processing (NLP).
 - Used in:
 - ❖ large-scale natural language processing
 - ❖ computer vision (vision transformers)
 - ❖ Reinforcement learning
 - ❖ Audio
 - ❖ multimodal learning
 - ❖ Robotics
 - Led to the development of pre-trained systems, such as generative pre-trained transformers (GPTs) and BERT (bidirectional encoder representations from transformers).
- Primarily used in the AI subfields of natural language processing (NLP) and computer vision (CV).

Transformer Architecture

Key Components

- Attention Mechanism: Enables the model to focus on relevant parts of the input.
- Self-Attention and Multi-Head Attention: Captures different aspects of the input simultaneously.
- Positional Encoding: Retains the order of input sequences.



Why Go Beyond NLP?

Unified Architecture: Transformers offer a generic framework adaptable across domains — from language to vision and beyond.

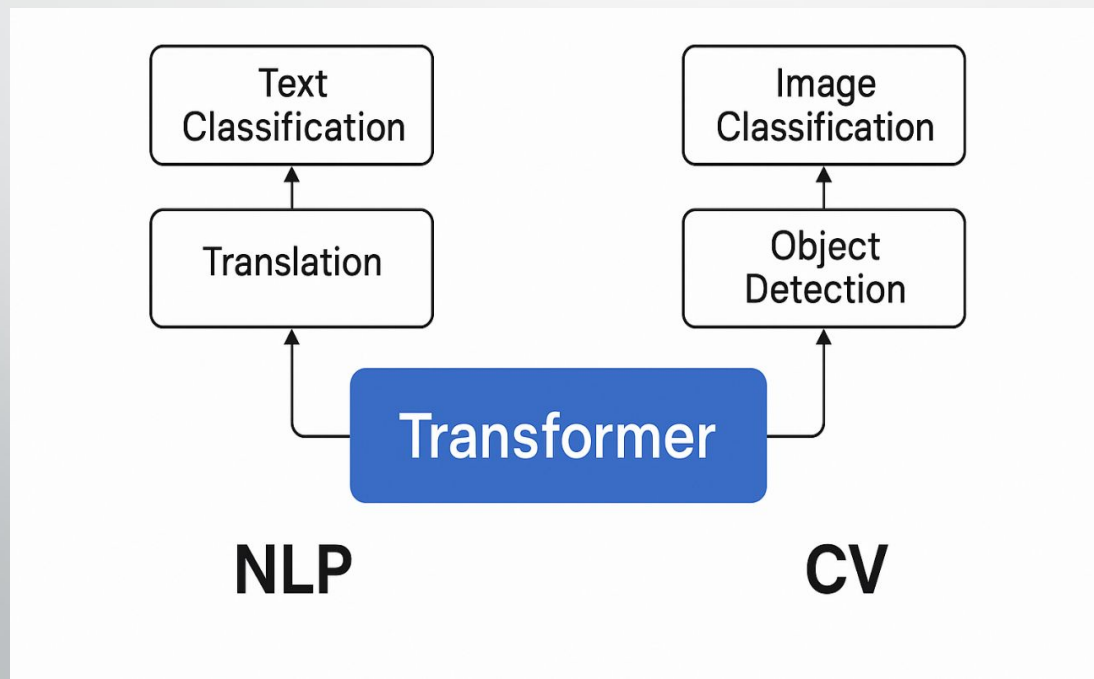
Success in NLP: Transformers like BERT and GPT have significantly outperformed previous models, inspiring cross-domain applications.

Limitations of CNNs: Convolutional Neural Networks struggle with modeling long-range dependencies and global context in images.

Transformer Strengths in Vision: Self-attention mechanisms allow Vision Transformers to capture global relationships in images effectively.

Scalability & Parallelism: Transformers process data in parallel, making them more efficient on modern hardware compared to sequential models.

Data-Driven Performance: With sufficient training data, ViTs can match or surpass CNNs in key computer vision benchmarks.



From Text to Vision – The Evolution

- Vision Transformers(ViTs) has been introduced that extends the transformers to Computer vision tasks.
- Supervise-trained to predict the image label from the patches of the image.
- Alternative to Convolutional Neural Networks (CNNs) that are currently state-of-the-art (SOTA) in different image recognition computer vision tasks.
- ViT models outperform the current SOTA CNNs by almost x4 in terms of computational efficiency and accuracy.

Vision Transformers

ViT Architecture



How it works?

The overall structure of the vision transformer architecture consists of the following steps:

1. Split an image into patches (fixed sizes)
 2. Flatten the image patches
 3. Create lower-dimensional linear embeddings from these flattened image patches
 4. Include positional embeddings
 5. Feed the sequence as an input to a SOTA transformer encoder
 6. Pre-train the ViT model with image labels, then fully supervised on a big dataset
- Fine-tune the downstream dataset for image classification

Real-World ViT Use Cases and Applications

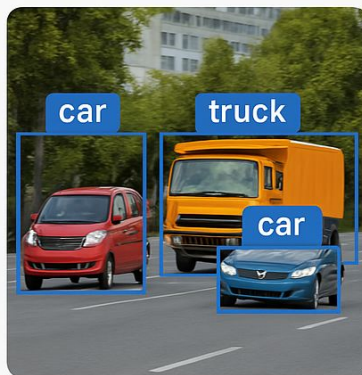
- **Image Classification:** ViT has shown excellent performance on large-scale image classification tasks like ImageNet.
- **Object Detection:** Transformer-based models like **DETR** (Detection Transformer) are being used for object detection tasks, improving accuracy by capturing long-range dependencies between objects in images.
- **Image Segmentation:** Transformers are also being used in medical imaging for tasks like tumor segmentation.

Real-World ViT Use Cases and Applications

Applications of Vision Transformers



Image
Classification



Object
Detection



Segmentation

Real-World ViT Use Cases and Applications



Real-World ViT Use Cases and Applications



ViT vs. CNN

Feature	Vision Transformer (ViT)	Convolutional Neural Network (CNN)
Feature Extraction	Global, using non-overlapping image patches	Local, through convolutional filters
Architecture	Based on transformers	Built around convolutional layers
Data Requirement	Requires more data to perform optimally	Less data-intensive, can perform well on smaller datasets
Computational Cost	Higher, due to the need for large-scale pre-training	Relatively lower
Use Case Suitability	Excels in tasks requiring an understanding of global patterns	Ideal for tasks where local features are significant

Strengths of Vision Transformer Architecture

- **Global Context:** ViTs are excellent at capturing long-range dependencies between image patches, giving them a better understanding of the global context of images.
- **Scalability:** ViTs scale well with larger datasets and deeper architectures, making them highly effective for large-scale vision tasks.
- **Flexibility:** Since ViTs don't rely on convolutions, they can be easily adapted to different tasks and domains beyond image classification, such as video analysis and object detection.

Challenges and Limitations

- **Data Requirements:** ViTs are data-hungry and typically require large-scale datasets for effective training. Fine-tuning ViTs pre-trained on large datasets (like ImageNet) is a common practice when data is limited.
- **Computational Resources:** Vision Transformers demand significant computational power, especially during training, due to the quadratic complexity of the self-attention mechanism.

Conclusion

- **Transformers have redefined deep learning**, starting from NLP and now reshaping the landscape of computer vision through Vision Transformers (ViTs).
- **ViTs demonstrate that attention mechanisms**, when applied to visual data, can match or exceed the performance of traditional Convolutional Neural Networks (CNNs), particularly when trained on large-scale datasets.
- Despite their **computational demands and data requirements**, ViTs offer a flexible, scalable, and architecture-agnostic framework that opens new frontiers across modalities.
- Ongoing research into **data-efficient training, hybrid models, and lightweight ViTs** is rapidly addressing their current limitations, making them increasingly viable for real-world applications.
- As deep learning continues to evolve, **transformers are poised to become a unifying model across domains**, enabling more robust, generalizable, and intelligent AI systems.



The End